

Michael Pignatelli

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EDUCATION

University of Pennsylvania

Philadelphia, PA

Bachelor of Science in Engineering in Computer Science and Cognitive Science (Dual Major) *Expected May 2027*

- Minor in Italian Culture | GPA: 3.56/4 | Affiliations: Claude Builder Club, Wharton Data Science Group

Master of Science in Engineering in Computer and Information Science (Sub-matriculation) *Expected August 2027*

- Concentration in Artificial Intelligence | GPA: 3.85/4 | All synchronous coursework complete by May 2027.

Università di Bologna | *Exchange Student – Bologna Consortial Studies Program*

January 2026 – June 2026

- Coursework in NLP, Cognitive Neuroscience, and advanced Italian language & culture at the world's oldest university.

PROFESSIONAL EXPERIENCE

Bentley Systems | *Software Development Engineering Intern*

June 2026 – August 2026

- Integrated Bentley Copilot, an AI assistant, into the SACS Offshore and iTwin Studio platforms to enable AI-driven iModel report generation and analysis.
- Collaborated with a fellow intern to build MCP (Model Context Protocol) server integrations connecting SACS Offshore DB and other analysis tools to Bentley Copilot.
- Enabled secure AI tool integration for Bentley Copilot by implementing required Azure AD authentication.

Ava Labs | *Software Engineering Intern – Platform Team*

June 2025 – August 2025

- Implemented and productionized 4 gRPC APIs (+7.5k LoC) in the AvalancheGo blockchain client, replacing legacy JSON-RPC and enabling strongly-typed communication for all node operators.
- Reduced parsing errors and improved type safety, delivering an expected ~50% lower latency vs REST for ~1,800 clients.
- Wrote end-to-end and integration tests (Ginkgo) to validate gRPC API adoption and ensure cross-version stability.
- Authored incremental deployment plan adopted for long-term rollout.
- Collaborated with Data Tooling and Cloudflare teams to align gRPC rollout with multi-team infrastructure standards.

Scale AI (via Outlier AI) | *Prompt Engineer (Contract)*

May 2024 – August 2024

- Engineered and validated 500+ prompts for LLM fine-tuning on math reasoning and algorithmic coding tasks, improving model accuracy and dataset consistency.
- Reviewed peer prompts and enforced quality standards, ensuring consistency across datasets and difficulty levels.

LEADERSHIP & TECHNICAL EXPERIENCE

University of Pennsylvania | *Head Teaching Assistant – Computer Science (CIS 1100)*

April 2025 – Present

- Lead and mentor 40+ undergraduate and graduate TAs for Penn's largest intro CS course (~300 students), overseeing recitations, office hours, and grading; maintained TA quality rating of 3.41/4 – well above CIS dept. average.
- Develop and deliver instruction on data structures, recursion, OOP, functional programming, and Pandas.
- Maintain course infrastructure (cis1100.com) with HTML/YAML/CSS, reducing site errors and ensuring clarity.
- Collaborate with faculty to update curriculum, author exams, and lead weekly staff meetings.
- *Teaching Assistant (August 2024 – April 2025)*: Supported course infrastructure and student instruction under Head TAs before being promoted to lead the team.

Penn High Powered Rocketry | *Engineering Mentor – Avionics, Recovery & Payload*

August 2026 – Present

- Advise underclassmen leads across avionics, recovery, and payload subsystems on design and testing, continuing the club's tradition of transitioning ownership after a lead year.
- *Recovery Engineering Lead (May 2025 – August 2026)*: Led recovery system design for L3 rocket; despite an in-flight CATO from a defective COTS motor, the rocket recovered successfully, earning judges' commendation for its resilience.
- *Recovery Engineer (September 2023 – May 2025)*: Simulated flight dynamics (<5% error vs actual) supporting a successful 12,000 ft launch at Spaceport America Cup/IREC; integrated telemetry testing to ensure recovery hardware redundancy.

SELECTED PROJECTS

Tabby — FastAPI | MongoDB | React | Google Gemini Vision API

- Built full-stack bill-splitting app parsing receipts via Google Gemini Vision API to automatically split bills among users.
- Implemented proportional tax/tip distribution, direct Venmo integration, and real-time payment tracking; demoed to 500+ peers at PennApps Hackathon under a 36-hour build deadline.

PennOS — C | Operating Systems

- Built a Unix-like operating system implementing process management, scheduling, signal handling, and a custom shell.
- Served as primary contributor to the PennFAT file system, the project's core storage subsystem design and implementation.

Lifeline — React | TypeScript | TailwindCSS | Gemini API | [GitHub]

- Designed and built an accessible healthcare management web app for CIS 4120 (HCI), organizing medications, lab results, and appointments into a single timeline.
- Led user research, prototyping, and heuristic evaluation; implemented the core UI and Gemini API integration for AI-assisted features.

Pedestrian Buddies — TensorFlow | TensorFlow Lite | Arduino | [GitHub]

- Built an embedded CV pipeline to detect crossing-guard gestures in real time and dynamically control traffic signals.
- Optimized inference for hardware-constrained systems and deployed the model on Arduino using TensorFlow Lite.

SKILLS

Programming & Frontend: Python, Go, Java, C/C++, OCaml, SQL, Swift, JS/TS, React, SwiftUI, HTML/CSS, Figma

Backend/Infra: Docker/Containerization, Git/GitHub (Actions), CI/CD, Testing (JUnit, Ginkgo, pytest), Linux, gRPC/Protobuf, REST, FastAPI, MongoDB, Cloudflare

ML/Data Science: NLP, RAG, MCP, Fine-Tuning, Prompt Engineering, TensorFlow/PyTorch/Keras, Pandas/NumPy

APIs/Tools: Google Gemini Code Assist, OpenAI/Codex, Anthropic/Claude Code, Jupyter, XML/JSON